

KH26-ME-26 – Digital Twin-Based Predictive Maintenance and Machine Health Management

Background

Industrial machines may fail because of vibration, overheating, misalignment, lubrication problems or component wear. Conventional maintenance based only on fixed schedules can lead to unnecessary servicing or unexpected breakdowns.

Problem Statement

Develop an AI-assisted digital twin platform that represents the operating condition of an industrial machine using vibration, temperature, speed, load and maintenance-history data. The system should estimate machine health, detect abnormal behaviour and predict possible component failure. It should recommend inspection or maintenance actions based on risk level and remaining useful life. Operators must be able to compare current and historical performance through a dashboard.

Mandatory Requirements

- Machine-parameter monitoring
- Digital representation of machine condition
- AI-based anomaly and failure prediction
- Machine-health score
- Maintenance recommendations
- Historical trend visualization

Expected Prototype

A working digital-twin dashboard demonstrating machine-health analysis and maintenance prediction using simulated or sensor-generated data.

KH26-ME-27 – AI-Based Manufacturing Quality Inspection and Defect Traceability System

Background

Manual inspection of manufactured components is time-consuming and may produce inconsistent results. Identifying the cause of repeated defects is also difficult when production and inspection information are stored separately.

Problem Statement

Develop an AI-powered quality-inspection system that detects surface defects, dimensional variations or assembly errors from product images or sensor data. The platform should

classify defect type, estimate severity and associate the defect with machine, batch or production-stage information. It should generate alerts when defect rates exceed acceptable limits and display trends that help engineers identify possible root causes.

Mandatory Requirements

- Image or sensor-based defect detection
- Defect classification and severity scoring
- Batch and machine traceability
- Quality-threshold alerts
- Defect-trend analysis
- Inspection report generation

Expected Prototype

A functional inspection platform demonstrating defect detection and traceability using sample component images or simulated production data.

KH26-ME-28 – Intelligent Production Planning and Energy Optimization System

Background

Manufacturing units must meet production targets while managing machine availability, energy consumption, worker shifts and delivery deadlines. Poor scheduling can result in idle machines, increased power costs and delayed orders.

Problem Statement

Develop an AI-assisted production-planning platform that schedules jobs across available machines based on processing time, machine capacity, energy demand and delivery priority. The system should minimize idle time, peak-hour electricity usage and production delays. It must respond to machine breakdowns or urgent orders by generating an updated schedule. Managers should be able to compare alternative production plans through a dashboard.

Mandatory Requirements

- Job and machine-data management
- AI-based production scheduling
- Energy-aware machine allocation
- Breakdown and urgent-order rescheduling
- Delay and utilization analysis
- Production monitoring dashboard

Expected Prototype

A simulation-based application demonstrating optimized production scheduling across multiple machines and jobs.

KH26-ME-29 – AI-Assisted Agricultural Machinery for Precision Field Operations

Background

Small and medium farmers often depend on manual observation for irrigation, spraying and crop monitoring. Uniform application of water or chemicals can increase cost, waste resources and damage healthy crops.

Problem Statement

Develop an intelligent agricultural-machine prototype that uses field images, sensor readings or location data to support selective irrigation, spraying or crop monitoring. The system should identify areas requiring attention and recommend or control the required operation. It should maintain activity records, estimate resource consumption and allow manual control by the farmer. The design should be affordable and suitable for small farms.

Mandatory Requirements

- Crop or field-condition monitoring
- AI-based treatment-zone identification
- Selective irrigation or spraying control
- Resource-usage calculation
- Manual override facility
- Farmer-friendly monitoring interface

Expected Prototype

A hardware model or simulation demonstrating zone-based agricultural operation using sample field data.

KH26-ME-30 – Smart Material Recovery and Robotic Waste Segregation System

Background

Mixed municipal and industrial waste contains recyclable materials such as metal, plastic, paper and glass. Manual segregation is slow, unsafe and may fail to recover valuable materials efficiently.

Problem Statement

Develop an AI-assisted waste-segregation system that identifies material categories using images, sensors or physical properties. The system should direct each item to an appropriate collection section through a robotic arm, conveyor mechanism or simulated actuator. It should record category-wise waste quantities, identify uncertain objects and estimate the amount of recoverable material. Operators should be able to monitor system performance and manually correct incorrect classifications.

Mandatory Requirements

- Waste-object detection and classification
- Mechanical or simulated segregation mechanism
- Category-wise quantity tracking
- Uncertain-object handling
- Manual correction option
- Recovery and performance dashboard

Expected Prototype

A functional hardware or simulation-based prototype demonstrating automated identification, segregation and waste-recovery analytics.